YAGSL Eight Steps Debugging Process ni artham cheskovadam

FRC lo, swerve drive systems chaala powerful matrame kaakunda, sarigga configure cheyadaniki chaala kashtamaina drivetrains lo okati. Yet Another Generic Swerve Library (YAGSL) swerve programming ni sulabham cheyadaniki sahayapaduthundi. Ayinappatiki, teams inka motor inversions, encoder directions, gyro orientation, mariyu field-oriented control ki sambandhinchina samasyalani edurukovachu.

Vishaya Suchika

Swerve drives enduku fail avutayi?

Swerve drives chaala varaku coordinate consistency meeda aadharapadi untayi.

YAGSL ivanni nijamani anukuntundi:

- Wheel munduki thiragadam (positive wheel rotation) ante munduki velladam ani artham.
- Positive steering rotation ante counterclockwise direction lo thiragadam.
- Gyro angle sarigga perugutundi.
- Field-oriented transformations ganitha paranganga consistent ga untayi.
- Encoder readings nijamaina physical movement tho match avutayi.

Ivi emaina sarigga lekapothe, robot ilaga cheyavachu:

- Uncontrollable ga spin avvadam
- Pakkaki drift avvadam
- Translate avutunnappudu rotate avvadam
- Robot heading ni batti translational axes maradam
- Venakaki velladam
- Oscillate avvadam
- Autonomous paths fail avvadam

YAGSL developers “Enimidi adugulu” (Eight Steps) ane oka systematic troubleshooting process ni create chesaru. Idi FRC swerve programming lo chaala mukhyamaina debugging tools lo okati.

Ee document ila chebutondi:

- YAGSL Enimidi Adugulu ante enti
- Ee process enduku undi
- Inversion samasyala venuka unna theory
- Prathi step em marustundi
- Prathi configuration ni surakshithanga ela test cheyali
- Teams chese samanya thappulu
- Long-term reliability kosam uttama padhatulu

Ee document teams Enimidi Adugulani follow avvadaniki kavalsina vivaranalu mariyu uttama padhatulani vivarinstundi.

Enimidi Adugulu prarambhinchadaniki mundu

Samasya mechanical ga vachindi kaadu ani dhrudaparuchukovadaniki, teams kinda vatini verify cheyali:

- CAN IDs sarigga unnayi
- Hardware clients lo motors sarigga respond avutunnayi
- Absolute encoders connect ayyi unnayi
- Modules physically free ga thirugutunnayi
- Encoder magnets secure ga unnayi
- Battery charge ayyi undi
- Wheels sarigga install ayyi unnayi
- Gear ratios sarigga unnayi

Software checklist:

- Vendor libraries install ayyi unnayi
- JSON configuration files unnayi
- Sarigga module positions assign ayyi unnayi
- Gyro connect ayyi detect ayyindi
- Conversion factors sarigga calculate ayyayi

Aaru configurations sarigga pani cheyavani, okka configuration daggaraga correct ga anipistundani, mariyu okka configuration matrame poorthiga correct ga untundani YAGSL documentation hecharistundi. Thappudu configurations meeku mariyu mee robot ki hanikaram kaavachu.

Anduke test cheyadaniki mundu teams kinda vatini dhrudaparuchukovali:

- Modatilo robot ni blocks meeda pettandi
- Wheels ki dooranga undandi
- Emergency disable ready ga unchukoṇḍi
- Slow ga test cheyandi
- Debugging appudu low speed limits

Inversion ni artham cheskovadam

Motor Inversion

Motor inversion anedi motor output yokka sign (+ or -) ni marustundi.

Inversion false aithe, positive voltage munduki thirugutundi.

Inversion true aithe, positive voltage venakaki thirugutundi.

YAGSL lo, drive motor inversion prathi module JSON file lo configure cheyabadutundi.

Example:

```
"drive": {
   "invert": true
}
```

IMU (Gyroscope/Gyro) Inversion

Gyro robot yokka rotation ni kolustundi.

Gyro counterclockwise badulu clockwise perigithe, field-oriented control ganitha paranganga (mathematically) inconsistent ga marutundi.

YAGSL teams ki swervedrive.json lo kindi vidhanga gyro readings ni invert cheyadaniki anumathi istundi.

```
"invertIMU": true
```

Field-oriented control joystick inputs ni ivatiki madhya transform chestundi:

- Robot-relative coordinates
- Field-relative coordinates

Ee transformation poorthiga gyro orientation meeda aadharapadi untundi.

Oka vela gyro sign thappuga unte, transformation thappuga rotate avutundi.

Enimidi Adugulu

Enimidi Adugulu anevi saadyamaina inversion combinations ni structured ga vethakadam.

Ikkada rendu mukhya variables unnayi:

1. IMU inversion
2. Drive motor inversion

Module orientation ni flip cheyalsi raavachu.

Oka combination sarigga pani chese varaku ee process systematic ga prathi combination ni test chestundi.

Step 1

Baseline configuration ni set cheyadaniki, ilaa pettandi:

```
invertIMU = false
```

Anni drive motor inversions = false ga pettandi

Ee samayam lo:

- Gyro normal ga undi ani anukuntamu
- Drive motors normal ga unnayi ani anukuntamu

Robot ni field-oriented drive upayoginchi test cheyandi.

Saadyamaina phalitalu:

- Robot gundranga violent ga thirugutundi
- Translation axes thappuga marutayi
- Robot venakaki veltundi
- Steering unstable ga kanipistundi

Mee drive problem fix aithe, meeru ikkade aagipovachu.

Aithe, chaala teams ki Step 1 lo success raadhu. Ilaa jarigithe, kangarupadakandi; basica ga ee process ni munduku konasaaginchandi.

<u>Step 2</u>

Marchandi:

```
invertIMU = true
```

Drive motors uninverted gaane untayi.

Ippudu gyro direction flip ayyindi.

Physical mounting, firmware conventions, mariyu coordinate systems ni batti konni gyros clockwise perugutayi, inkakonni counterclockwise perugutayi. Anduke ee step avasaram.

Field-oriented motion sudden ga improve aithe, gyro inversion e samasya ayundavachu.

Common Improvement
Translation ippudu chaala varaku pani chestundani, rotation melugga undani, mariyu drift thagginatluga teams gamanisthu untaru. Kaani robot inka vichitramga behave chesthe, meeru munduku konasaagali.

<u>Step 3</u>

Marchandi:
Anni drive motors ni invert cheyandi

Meeru ee maarpulani chustaru:

```
invertIMU = true
```

Drive inversions = true

Enduku?
- Konni modules mechanically wheel direction ni reverse chestayi.
- Steering sarigga pani chesina, wheel direction software expectations ki vyethirekanga undavachu.

Deeni valla ilaa jaragavachu:
- Axis swapping
- Heading-based inversion
- Unstable odometry

Robot sudden ga normal ga driven aithe, mee drive motors ki inversion avasaram ayundavachu.

Paristhithi inka ekkuva paadaithe, thadupari step ki vellandi.

Step 4

Marchandi:
```
invertIMU = false
```
Drive motors inverted gaane untayi.

Idi deeni test chestundi:
- Inverted drive motors
- Normal gyro orientation

Ee samayaniki robot ilanti combinations anni complete chesindi:
- Gyro normal/inverted
- Drive motors normal/inverted

Eevi sarigga pani cheyaka pothe, samasya physical module orientation ki sambandhinchindi ayundavachu.

Step 5

Marchandi:

Modules ni flip cheyandi (software definitions ki anugunanga module positions ni swap cheyandi)

Udaharana ki:
- Front left anedi back right avutundi
- Front right anedi back left avutundi

Idi ilaa cheyavachu:
- JSON files ni rename cheyadam dwara
- Module coordinates ni swap cheyadam dwara
- CAN configurations ni reassign cheyadam dwara

Idi enduku sahayapadutundi:
- Swerve kinematics konni nirdishta module locations ni anukuntundi.
- Oka module configure chesina daani kante physical ga vereru ga mount aithe, robot yokka mathematical model thappuga marutundi.

- Idi rotational instability, translation drift, mariyu thappudu wheel optimization ki kaaranam kaavachu.

Modules ni flip cheyadam valla coordinate mismatches sariavutayi.

<u>Step 6</u>

Marchandi:
Drive inversions ni thirigi false ki set cheyandi

Ippudu,
- Modules flip ayyi unnayi
- IMU normal ga undi
- Drive motors normal ga unnayi

Idi kotha module arrangement tho inversion search ni malli repeat chestundi.

<u>Step 7</u>

Marchandi:
invertIMU = true
Drive motors uninverted gaane untayi.

Idi inverted gyro orientation tho flipped module layout ni test chestundi.

Chaala teams ki ikkada success dhorukutundi.

<u>Step 8</u>

Marchandi:
Drive motors ni malli invert cheyandi

Ippudu:
Modules flip ayyi unnayi
IMU inverted ga undi
Drive motors inverted ga unnayi

Idi chivari inversion combination.

Step 8 tharvatha kooda robot thappuga behave chesthe, samasya sadharananga inversion ki sambandhinchindi kaadu.

Oka vela enimidi adugulu fail aithe:

Samasya chaala varaku drive motors mariyu IMU inversion ki baita hone ki avakaasam undi. Teams kinda vatini pariseelinchali.

<u>Hardware Problems</u>
- Thappudu CAN IDs
- Paadaina encoders
- Loose ga unna magnets

- Damaged motors
- Thappudu gearing
- Thappudu module assembly

Configuration Problems
- Thappudu wheel diameter
- Thappudu conversion factors
- Thappudu encoder offsets
- Thappudu PID values
- Thappudu module positions

Mechanical Problems
- Vankaraina shafts
- Binding gears
- Loose ga unna belts
- Slip avutunna encoder magnets

Software Problems
- Thappudu joystick mapping
- Thappudu coordinate conventions
- Mixed vendor library versions
- Outdated YAGSL installation

Absolute encoder offsets

Swerve instability ki chaala samanyamaina kaaranallo thappudu absolute encoder offsets okati.

Robot boot aynappudu prathi wheel ki daani exact orientation theliyali.

Offsets ni configure cheyadaniki:
- Anni wheels ni munduki petti align cheyandi.
- Encoder values ni read cheyandi.
- Offsets ni module JSON files lo store cheyandi.
- Reboot tharvatha kooda wheels aligned ga unnayo ledo verify cheyandi.

Thappudu offsets inversion problems laga anipinchavachu.

Symptoms:
- Wheels thamanu thamu eppudoo correct cheskuntu undadam
- Rest lo unnappudu modules twitch avvadam
- Wheels veru veru directions lo point ayyi undadam
- Robot pakkaki drift avvadam

PID tuning

Konni sarlu asalu samasya PID tuning ayinaa, teams thappuga inversion ni blame chestuntaru.

Oka vela steering PID chaala aggressive ga unte:
- Modules oscillate avutayi
- Wheels jitter avutayi
- Robot vibrates avutundi

Oka vela steering PID chaala weak ga unte:
- Wheels commands ki venakabadi avutayi

- Robot sloppy ga anipistundi
- Thappudu drive PID

Thappudu drive PID valla ilaa jaragavachu:
- Jerky acceleration
- Poor autonomous accuracy
- Drifting

Enimidi Adugulu coordinate consistency ni solve cheyagalaru, kaani avi robot ni tune cheyavu.

Teams kosam uttama padhatulu

1. <u>Example code tho prarambhinchandi</u>

YAGSL yokka example project prathyekanga debugging time ni thagginchadaniki design cheyabadindi.

Robot basic TeleOp driving ni vijayavantanga complete cheyakundane pedda custom drive systems rayadam thappinchukondi. Chaala teams drivetrain sarigga pani chestundo ledo verify cheyakundane ventane advanced autonomous routines, vision integration, mariyu custom odometry ni add cheyadam prarambhistaru. Idi chaala sarlu isolate cheyadaniki kashtamaina multiple overlapping problems ni srushtistundi.

Deeniki badulu, teams ilaa prarambhinchali:
- Basic joystick driving
- Sarigga unna module steering
- Stable field-oriented control
- Reliable gyro readings
- Sariyaina encoder offsets

Robot eppudaithe consistent ga correct ga drive avutundo, appudu kotha systems ni koddikoddiga add cheyavachu.

Munde pani chestunna example code ni upayoginchadam valla samasya hardware, configuration, leda custom programming logic nunchi vachindo isolate cheyadaniki sahayapadutundi.

Oka vela example project pani chesi custom code pani cheyakapothe, samasya YAGSL lo kaakunda team chesina modifications lo unde avakaasam undi.

2. <u>Incremental testing upayoginchandi</u>

Motham robot functionality ni okkesari try cheyakunda, oka sari ki okka subsystem ni test cheyandi.

Sifarisu chesina kramam:
1. Steering motors sarigga rotate avutunnayo ledo verify cheyandi.
2. Drive motors munduki sarigga kaduluthunnayo ledo verify cheyandi.
3. Encoder readings sarigga update avutunnayo ledo confirm cheyandi.
4. Gyro orientation ni test cheyandi.
5. Robot-relative drive ni enable cheyandi.

6. Field-oriented drive ni enable cheyandi.
7. Odometry ni test cheyandi.
8. Chivariga autonomous path driven ni test cheyandi.

Ee structured approach debugging time ni chaala thaggistundi endukante samasyalani nirdishta subsystem ki isolate cheyavachu.

3. <u>Annitini document cheyandi</u>

Vijayavantamaina FRC teams thama drivetrain configuration ki sambandhinchina vivaralani detailed records ga meintain chestai.

Teams ivanni document cheyali:

- CAN IDs
- Encoder offsets
- Gear ratios
- Wheel diameter
- PID constants
- Inversion settings
- Module locations

Documentation lekapothe, chinna maarpulu kooda season lo tharvatha reverse cheyadam kashtamangaa marutundi.

Manchi documentation kotha students ki system nerchukovadaniki, programmers ki fast ga debug cheyadaniki, mariyu future seasons ki smooth ga start avvadaniki sahayapadutundi.

4. <u>Chala mandi vidyarthulaku shikshana ivvandi</u>

Swerve programming knowledge eppudoo okka student daggare aagipokoodadu.

Drivetrain knowledge kotha members ki transfer kaakapovadam valla, experienced programmers graduate ayinappudu chaala teams pedda setbacks ni edurkuntai.

Deenini thappinchadaniki:

- Kotha students ki Enimidi Adugulu (Eight Steps) ela pani chestayo nerpandi
- Coordinate systems mariyu inversion logic ni vivarinchandi
- Debugging sessions ni kalisi cheyandi
- Internal guides mariyu training presentations ni srushtinchandi

Knowledge ni sarigga share cheskune teams eppudoo long-term lo ekkuva success avutayi endukante avi okke expert meeda aadharapadakunda sustainable engineering practices ni nirmistayi.

Mee programming prayananiki all the best!